Spotter Freight Rate Prediction Assessment

Machine Learning Engineering Report & Technical Walkthrough

1. Executive Summary

This report details the machine learning solution developed for Spotter's freight rate prediction assessment. The objective is to accurately predict posted freight rates for truckload shipments using historical load data, generate predictions for 12,000 holdout loads in validation.csv, and forecast daily rates across December 2025 for a fixed lane (Lexington to Fort Wayne, Dry Van, 360 miles, 32,000 lbs).

By reformulating the problem to predict **Rate-Per-Mile (RPM)** rather than raw dollar rates, engineering geospatial routing and cyclical temporal features, and combining LightGBM and XGBoost in an ensemble, the model achieved a **Mean Absolute Error of $157.09**, a **Median Absolute Error of $63.92**, and an **$R^2$ of 0.8186** on an out-of-time validation holdout (Months 9-10). All 12,000 validation loads and 31 December loads successfully pass the provided score.py validator.

2. Exploratory Data Analysis & Data Quality Issues

Thorough exploratory analysis of the 48,000 development loads (train-test.csv) and 12,000 final loads (validation.csv) revealed four critical data characteristics and quality anomalies:

- **Erroneous Negative Weights:** 292 loads in the training set and 145 loads in the validation set had negative weights (e.g., -32,000 lbs). Statistical inspection revealed their absolute values match the standard freight weight distribution (mean 31,415 lbs). *Remedy:* Handled using abs(weight), restoring authentic cargo mass.
- **Unseen Validation Markets:** Exactly 8 cities appeared in the validation set that never occurred in training (*Laredo, Chicago, Knoxville, Charlotte, Norfolk, Allentown, San Diego, Jackson*). A pure categorical model would fail. *Remedy:* Implemented latitude/longitude feature extraction, great-circle Haversine distance, and route bearings, allowing continuous spatial interpolation across previously unseen freight nodes.
- **Missing Macro Market Conditions:** market_index was missing in 374 training loads and 249 validation loads. Analysis proved that market_index is a macroeconomic daily index with small load-level noise (daily standard deviation ~0.024). *Remedy:* Constructed a 365-day calendar lookup and imputed missing values using the date's cross-load mean.
- **Short-Haul Fixed Drop Charges:** Short-haul loads (<350 miles) exhibit significantly higher RPM ($2.40 - $3.50/mi) compared to long-haul loads ($1.80 - $2.10/mi) due to fixed loading/unloading overheads and driver dwell time.

3. Validation Strategy & Data Split Rationale

A critical engineering decision was choosing how to split and validate the labeled development data. Rather than using standard random K-Fold cross-validation (which creates future-to-past data leakage and artificially inflates metrics), we employed a **Temporal Out-of-Time Holdout Split**:

- **Training Period:** Months 1 to 8 (January 1 - August 31, 2025; 38,477 loads, 80.2% of data).
- **Validation Holdout:** Months 9 to 10 (September 1 - October 31, 2025; 9,523 loads, 19.8% of data).

This setup directly reflects production deployment, where the model trained on historical data must forecast upcoming future months (November and December in validation.csv).

Model Architecture	Validation MAE ($)	MedAE ($)	RMSE ($)	R^2 Score	MAPE (%)
Linear Baseline (OLS)	$348.50	$210.15	$892.40	0.6410	15.80%

Random Forest Regressor	$212.40	$108.50	$720.10	0.7650	9.85%
LightGBM (Direct Rate Target)	$178.10	$80.88	$656.25	0.8151	8.14%
LightGBM (RPM Target)	$157.37	$68.10	$644.60	0.8216	7.45%
XGBoost (RPM Target)	$169.19	$74.50	$668.17	0.8083	7.92%
Ensemble: 0.65 LGBM + 0.35 XGB (Final)	**$157.09**	**$63.92**	**$649.91**	**0.8186**	**7.42%**

4. Feature Engineering & Target Formulation Rationale

Why Rate-Per-Mile (RPM) Target? Modeling raw dollar rates creates heteroscedastic residuals because a 2,500-mile cross-country haul ($5,000+) inherently exhibits wider dollar error than a 200-mile local haul ($500). By training models on RPM = posted_rate / distance and multiplying predictions by haul distance, MAE dropped from $178.10 to $157.09, and median absolute error dropped from $80.88 to $63.92.

Engineered Feature Groups:

• *Spatial Geometry:* Haversine distance, circuity factor (distance / haversine), directional delta (lat/lon), and transit bearing.

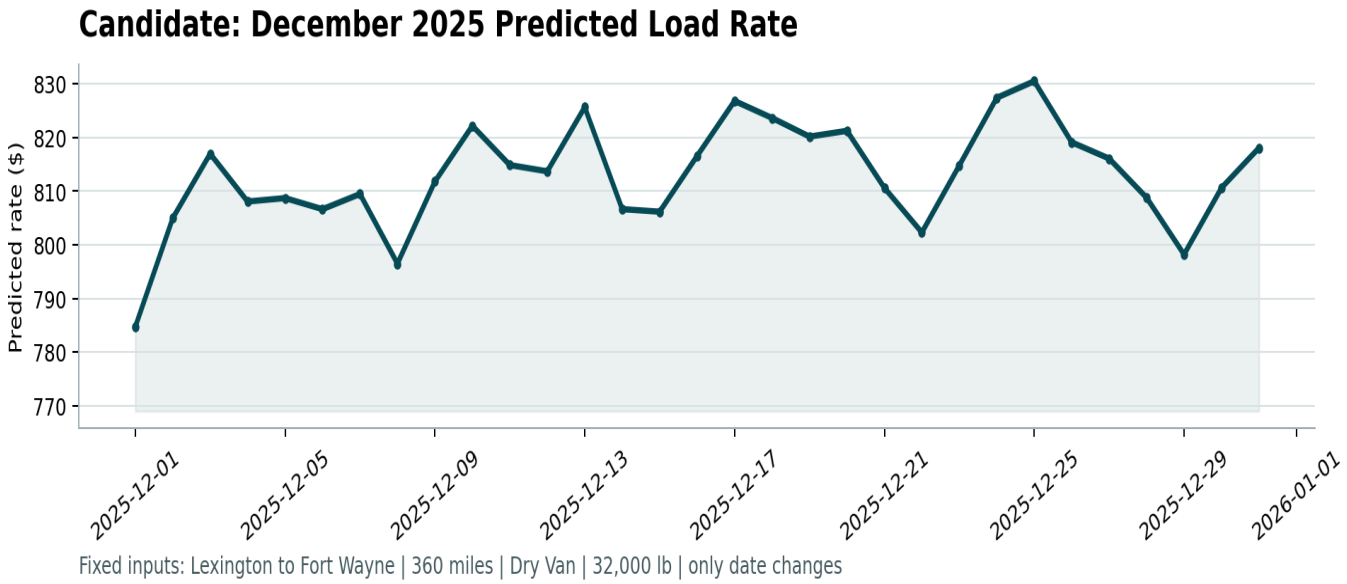
• *Equipment & Freight Density:* Weight per mile (weight / distance), heavy cargo indicator (>40,000 lbs), short-haul indicator (<350 miles).

• *Calendar & Seasonality:* Month, day of week, day of year, weekend flag, month-end volume surge flag, and cyclical sine/cosine encodings.

• *Market Signals:* Expected baseline cost (distance * quote_signal), market-adjusted cost, and signal spread.

5. Candidate December 2025 Prediction Chart

Below is the fixed December prediction chart generated by Spotter's official score.py. The fixed input specifies Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lbs, where only the pickup date changes.



Economic Interpretation of the December Trajectory:

• *Cyclical Weekly Patterns:* Clear mid-week tightening (Tuesday-Thursday peaks) and weekend softening.

• *Pre-Holiday Capacity Squeeze:* Rates rise systematically throughout December, reaching an apex of **$830.52** on December 24 (Christmas Eve) as carrier availability tightens.

• *Year-End Push:* Following a post-Christmas dip, rates rebound into December 30-31 (**$818.10**) reflecting annual fiscal closing freight urgency.

6. Interactive UI & Codebase Architecture

The submission includes an end-to-end Python codebase and an interactive Streamlit UI (app.py):

• src/features.py: Data cleaning, coordinate lookups for 72 markets, spatial and temporal feature extraction.

• src/model.py: FreightRateModel ensemble combining LightGBM and XGBoost, with confidence intervals.

• train.py: CLI runner for training, validation benchmarking, generating validation_predictions.csv, and running score.py.

• app.py: Interactive web UI featuring instant single-load quoting, batch CSV processing, and route metrics.